

REPORT CLASSIFICATION

Classification Type	Full Form Analysis
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DATE / TIME

Date of Occurrence	2012-07
Local Time Of Day	1801-2400

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Light	Night
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AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737-800
Mission	Passenger
Flight Phase	Parked
Maintenance Status - Maintenance Deferred	N
Maintenance Status - Released For Service	Y
Maintenance Status - Maintenance Type	Unscheduled Maintenance
Maintenance Status - Maintenance Items Involved	Installation
Maintenance Status - Maintenance Items Involved	Repair

COMPONENT 1

Aircraft Reference	X
Aircraft Component	Fuselage Skin
Manufacturer	Boeing

PERSON 1

Location Of Person	Gate / Ramp / Line
Reporter Organization	Air Carrier
Function - Maintenance	Technician
Qualification - Maintenance	Airframe
Qualification - Maintenance	Powerplant
Experience - Maintenance Technician	14 yrs
ASRS Report Number	1022467
Human Factors	Situational Awareness
Callback	Completed

EVENTS

Anomaly	Deviation / Discrepancy - Procedural - Maintenance
Anomaly	Deviation / Discrepancy - Procedural - Published Material / Policy
Detector - Person	Maintenance
Were Passengers Involved In Event	No
When Detected	Routine Inspection
When Detected	Aircraft In Service At Gate
Result - General	Maintenance Action

ASSESSMENTS

Contributing Factors / Situations	Human Factors
Contributing Factors / Situations	Manuals
Contributing Factors / Situations	Procedure
Primary Problem	Manuals

NARRATIVE 1

It has come to my attention that I may have inadvertently used a Structural Repair Manual (SRM) repair that is not applicable to the area that was repaired. I became aware of this a few days after completing the repairs after receiving a company Maintenance Alert. The Maintenance Alert was titled B737-700/800 SRM Allowable Damage and Repairs on Bilge Skins, (posted July 2012). The Maintenance Alert noted the following: the SRM's for both -700/-800 fleet types have always included a statement in SRM 53-30-01 and 53-60-01 stating that the Allowable Damage and Section-53 General Repairs may not be used in the bilge skin, however, the area comprising the bilge skin was not clearly defined. In the most recent Boeing SRM Revisions, the SRM's have been revised to clearly define the term Bilge Skin:

Section 43 Bilge Skin: Frame Stations (FS) FS 360-540, Stringer (S) S-24L to S-24R

Section 46 Bilge Skin: FS 727-887, S-23L to S-23R.

I realized after reading this Maintenance Alert that I had used the SRM 53-00-01 Allowable Damage limits and repair on our company B737-800 aircraft two days prior to the Maintenance Alert being posted. So the damage locations stuck in my mind and I realized that the repair locations that I had repaired fell into the locations in the Maintenance Alert. Due to the [fuselage] "bilge areas" not being clearly defined and to the placement of the 'Note' in the SRM 53-60-01 on Page 101, Figure 101, "REFER TO SRM 53-60-70 FOR THE FAIRING SKIN ALLOWABLE DAMAGE. THERE IS NO ALLOWABLE DAMAGE FOR BILGE SKIN IN THE STRUCTURAL REPAIR MANUAL AT THIS TIME".

At the time I made the repairs on the B737-800 aircraft, I was not aware of the exact location of the "bilge area" and since the above underlined sentence had followed the 'Note' for the fairing skin, I thought the bilge area was only the area located under the fairings. As of writing this letter I have contacted the following people to make them aware of this: my Lead Mechanic, my Duty Manager, my Station Manager, Company Maintenance Control and FAA/NTSB Company Liaison notified me that he would contact Company Lead Engineer to see what would need to be done in this matter.

CALLBACK 1

Reporter stated there were several scratches in the external skin of the lower fuselage aft of the Main Landing Gear (MLG) wheel wells, under the aft pit. He had burnished out the fuselage skin to remove the sharp edges of the scratches. But after reading the company Maintenance Alert two days later regarding the bilge skin areas and discussing with Engineering, a decision was made to have Inspection perform an Ultra Sound Non-Destructive Testing (NDT) for skin thickness loss and scratch damage depth. A SRM Skin Repair was required for one of the scratched areas.

SYNOPSIS

A Line Mechanic reports he decided to revisit the lower fuselage external skin scratches that he had previously burnished out on a B737-800 aircraft after reading a company Maintenance Alert about revisions in the Structural Repair Manual (SRM) that clarified skin areas considered to be "bilge skin".

